

LIZE SHAO

zgr3et@virginia.edu | slz2003.github.io

RESEARCH INTERESTS

My work focus on **trustworthy and verifiable code generation**, particularly **neurosymbolic approaches** that integrate generative models with formal methods, program analysis, and constraint reasoning. My goal is to build coding agents that can reason about specifications and generate software with **machine-checkable correctness and security guarantees**.

EDUCATION

University of Virginia

Ph.D. in Computer Science, advised by Prof. [Wenxi Wang](#)

2025 – Present

Charlottesville, VA

Rice University

Dual Degree in Mathematics and Computer Science, advised by Prof. [Xia Hu](#)

2021 – 2025

Houston, TX

- Rice Trustee Distinguished Scholarship (Top 1%) [\[link\]](#)

PUBLICATIONS

- [1] **Constrained Code Generation with Discrete Diffusion** [\[link\]](#)
NeurIPS 2026. The Fortieth Annual Conference on Neural Information Processing Systems
[Lize Shao](#), Michael Cardei, Zichen Xie, Ferdinando Fioretto, Wenxi Wang
- [2] **VeriContest: A Competitive-Programming Benchmark for Verifiable Code Generation** [\[link\]](#)
NeurIPS 2026. The Fortieth Annual Conference on Neural Information Processing Systems
Zichen Xie, Mrigank Pawagi, Yuxin Liu, Aaditi Rai, [Lize Shao](#), John Berberian Jr., Sicong Che, Wenxi Wang
- [3] **RFCScope: Detecting Logical Ambiguities in Internet Protocol Specifications** [\[link\]](#)
ASE 2025. The 40th IEEE/ACM International Conference on Automated Software Engineering
Mrigank Pawagi, [Lize Shao](#), Hyeonmin Lee, Yixin Sun, Wenxi Wang
- [4] **MQuAKE-Remastered: Multi-Hop Knowledge Editing Can Only Be Advanced with Reliable Evaluations** [\[link\]](#)
ICLR 2025. The Thirteenth International Conference on Learning Representations
Shaochen Zhong, Yifan Louie Lu, [Lize Shao](#), Bhargav Bhushanam, ..., Vipin Chaudhary, Xia Ben Hu
- [5] **Detecting Inconsistencies in Model Specifications with LLM-as-Verifier Reasoning** [\[link\]](#)
Preprint.
Zichen Xie, Mrigank Pawagi, [Lize Shao](#), Yang Hu, Wenxi Wang
- [6] **HoCoS: Hyperbolic Representation Towards Code Search** [\[link\]](#)
Preprint.
Xunzhu Tang, [Lize Shao](#), Yewei Song, Saad Ezzini, Haoye Tian, Jiechao Gao, Jacques Klein, Tegawendé F. Bissyandé

HONORS & AWARDS

- Rice University Trustee Distinguished Scholarship (Top 1%) [\[link\]](#) 2021 – 2025
- 2nd Place, Chevron Challenge, Rice Datathon [\[link\]](#) 2023
- USA Mathematical Olympiad Qualifier [\[link\]](#) 2020
- USACO, Gold Division [\[link\]](#) 2020

ACADEMIC SERVICE

Reviewer: ICLR 2027 | **Sub-reviewer:** ICSE 2027

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Rust, SQL | **Formal Methods:** Verus, Lean, Z3 | **Tools:** PyTorch, Git, LaTeX